

HARRISON SIMS

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Third-year Ph.D. student at the University of Michigan with a robust foundation in psychology, biomedical engineering, and mathematics, working to advance research in human-automation team hierarchical structures and information exchange processes as a NIOSH occupational safety engineering trainee.

Education

Bachelor's Degree: Biomedical Engineering and Psychology (BS), Johns Hopkins University: 3.94 GPA

Graduation With Honors: May 2024

Leadership: BMES President, Blue Key Society, Alpha Phi Omega Service Fraternity

Honors Societies: Tau Beta Pi (Engineering), Psi Chi (Psychology), Alpha Eta Mu Beta (Biomedical Engineering)

Master's Degree: Industrial and Operations Engineering, University of Michigan: 4.00 GPA

Leadership: HFES Student Chapter Vice-President, HFES Occupational Ergonomics Technical Group Student Representative, Prospective Student Mentor

Graduation: May 2026

PhD: Industrial and Operations Engineering, University of Michigan: 4.00 GPA

Leadership: HFES Student Chapter President, HFES Occupational Ergonomics Technical Group Student Representative, Prospective Student Mentor, Graduate Student Lead

Expected Graduation: May 2029

Relevant Classes: Systems and Controls, Nonlinear Dynamics, Industrial and Organizational Psychology, Probability, Occupational Health Psychology, Advanced Statistical Methods, Applied Statistics/Data Analysis, Monte Carlo Methods, Stochastic Processes, Discrete Event Simulation, Work Measurement, Human Error & Complex System Failure, Global Cultural Systems Engineering, Interprofessional Perspectives in Occupational Health and Safety, Dynamic Programming, Quantifying Human Motion, Safety Engineering Methods

Honors and Awards:

Cesar E. Chavez Foundation Scholar ~ 2025

NSF Graduate Student Research Fellowships Program (GRFP) Honorable Mention (1 of 8 nationally) ~ 2025

Hispanic Scholarship Fund Scholar ~ 2020–2026

Rackham Graduate Student Research Grant ~ 2024

Achievement Rewards for College Scientists Scholar Awardee (ARCS) ~ 2023

Publications:

Sims, H., Ganberg, G., McCormack, R., & Volkova, S. (2025). ADCA: Artifact-Based Dataset Creativity Assessment. In *NeurIPS 2025 Workshop on Evaluating the Evolving LLM Lifecycle: Benchmarks, Emergent Abilities, and Scaling*.

Sims, H., & Liu, Y. (2025). A Quantitative and Empirical Study of the Dimensions of Teamness. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 0(0). <https://doi.org/10.1177/10711813251366311>.

Sims, H., Bhagat, I., & Liu, Y. (2025). The Characteristics of Learning Teams: A Quantitative and Empirical Study of Teamness in Education. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 0(0). <https://doi.org/10.1177/10711813251366299>

Sims, H., Bhagat, I., & Liu, Y. (2025). The Characteristics of Medical Teams: A Quantitative and Empirical Study of Teamness in Healthcare. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 0(0).

<https://doi.org/10.1177/10711813251357886>

- Sims, H.** (2025, September). The Development of a Novel Resilience Scorecard: What We Can Learn from Supply Chain Logistics. In *Proceedings of the International Symposium on Human Factors and Ergonomics in Health Care* (Vol. 14, No. 1, pp. 181-185). Sage CA: Los Angeles, CA: SAGE Publications.
- Sims, H.**, Neyens, D., Catchpole, K., Biro, J., Lusk, C., & Abernathy III, J. (2024). The Impact of a Novel Syringe Organizational Hub on Operating Room Workflow During a Surgical Case. *Joint Commission Journal on Quality and Patient Safety*.
- Sims, H.**, Biro, J., Lusk, C., Neyens, D. M., Catchpole, K., Shaik, A. S., ... & Abernathy III, J. H. (2023). The design and evaluation of a syringe hub for organizing syringes in anesthesia medication delivery. *Human Factors in Healthcare*, 4 100049.

Presentations:

- Sims, H.**, & Liu, Y. (2025). A Quantitative and Empirical Study of the Dimensions of Teamness. Human Factors and Ergonomics Society Annual Meeting 2025, October 13- October 17, 2025. Chicago, IL.
- Sims, H.**, Bhagat, I., & Liu, Y. (2025). The Characteristics of Medical Teams: A Quantitative and Empirical Study of Teamness in Healthcare. Human Factors and Ergonomics Society Annual Meeting 2025, October 13- October 17, 2025. Chicago, IL.
- Sims, H.**, Bhagat, I., & Liu, Y. (2025). The Characteristics of Learning Teams: A Quantitative and Empirical Study of Teamness in Education. Human Factors and Ergonomics Society Annual Meeting 2025, October 13- October 17, 2025. Chicago, IL.
- Islam, R., Huang, L., Willett, M., Teo, A., Klinefelter, Z., Bautista, P., Rajib, R., Reynolds, J., **Sims, H.**, Cohen, M., Cooke, N., & Fouse, A. (2025). Evaluating Human-Centered AI in Human–AI and Human–Human Teams Using Performance Metrics. Human Factors and Ergonomics Society Annual Meeting 2025, October 13- October 17, 2025. Chicago, IL.
- Sims, H.**, (2025). The Development of a Novel Resilience Scorecard: What We Can Learn from Supply Chain Logistics. International Symposium on Human Factors and Ergonomics in Healthcare, March 30- April 2, 2025. Toronto, ON
- Sims, H.**, Rosen, M., & Koka, R. (2024). Building the Connection Between Human Factors Analysis and Practical Application: A Case Study On The Complexities of Anesthesia Scheduling. International Symposium on Human Factors and Ergonomics in Healthcare, March 24-27, 2024. Chicago, IL
- Sims, H.**, Biro, J., Shaik, A., Rastogi, S., Neyens, D. M., Abernathy III, J.H., Catchpole, K.R., Lusk, C., & Woodward, J. (2023). The Perceptions of a Novel Syringe Organizational Device Following Implementation and its Impact on Operating Room Workflow During a Surgical Case. International Symposium on Human Factors and Ergonomics in Healthcare, March 26-29, 2023. Orlando, FL
- Woodward, J., Catchpole, K.R., Neyens, D. M., **Sims, H.**, Biro, J., Segarra, G., Lusk, C., & Abernathy III, J.H. (2023). Creation of a SEIPS 101 Tasks and Tools Matrix on Anesthesia Medication Delivery. International Symposium on Human Factors and Ergonomics in Healthcare, March 26-29, 2023. Orlando, FL
- Biro, J., Abernathy III, J.H., Catchpole, K.R., Lusk, C., Munie, S., Neyens, D.M., **Sims, H.**, & Wang, C. (2022). Technology acceptance across systems and roles: A survey of anesthesia providers evaluating a device designed to improve syringe organization during a surgical case. International Symposium on Human Factors and Ergonomics in Healthcare, March 26-29, 2022. New Orleans, LA

Technical Skills

Human Factors Tools: TAM, CTA, HTA, FRAM

Code: Python, MATLAB, Java, C++, R, Javascript

3D Printers: Makerbot, Stratasys, Ultimaker

Machine Learning: CLIP, BERT, BLIP, spaCy,
Hugging Face Transformers, embedding analysis

Development: HTML, CSS, Django, Flask, Shiny Apps

UX: Redcap, Zendesk, Figma, Qualtrics, Wireframes

CAD: Solidworks (CSWP), Onshape, Autodesk

Simulation: MCMC, Bootstrap, Kalman Filtering, DES

R Packages: lme4, ggplot2, tidyverse, psych

Work Experience

Warden Lab Manager, January 2026-Present ~ **University of Michigan**, Ann Arbor, MI

- Coordinated day-to-day laboratory operations, ensuring alignment of research activities, resources, and personnel across multiple ongoing projects.
- Organized and facilitated weekly lab meetings, setting agendas, tracking action items, and supporting clear communication between team members and the PI.
- Supervised and mentored undergraduate research assistants, delegating tasks, providing training, and monitoring progress to ensure high-quality contributions.

Intern Scientist, May 2025-August 2025 ~ **Aptima Inc.**, Woburn, MA

- Conducted advanced statistical analyses for DARPA-funded projects, applying ANOVA and linear mixed-effects models in R to extract key findings and generate visualizations.
- Performed a state-of-the-art literature review on evaluation methodologies for multimodal content generation—including text, images, videos, and time series—synthesizing findings to guide project design.
- Designed and implemented a modular Python-based evaluation pipeline for multimodal AI generation, integrating BLIP, CLIP, BERT, spaCy, and embedding similarity analysis to assess coherence, creativity, and consistency.
- Generated a synthetic image dataset with Kandinsky to support evaluation tasks and future research applications.
- Authored research abstracts, reports, and documentation to communicate results and ensure continuity of developed methods and tools.

Research Assistant, May 2024-August 2024 ~ **University of Michigan**, Ann Arbor, MI

- Conducted an in-depth literature review on human-human and human-robot team dynamics, systematically analyzing 50+ peer-reviewed articles to establish the theoretical foundation for doctoral research.
- Produced detailed research notes and synthesized key findings into monthly presentations for faculty advisor, guiding the refinement of dissertation research questions.
- Engaged in professional development through National Academies of Sciences, Engineering, and Medicine webinars on AI development and responsible oversight, integrating broader perspectives into research framing.

Human Factors Researcher, August 2022-May 2024 ~ **Armstrong Institute of Patient Safety**, Baltimore, MD

- Conducted research on the construction of anesthesia teams, specifically researching optimized schedule development.
- Designed an interview protocol, developed an interview script based on the functional resonance analysis method, and conducted several in-depth interviews to better understand the cognitive complexities of schedule development.
- Wrote an IRB application for sensitive data acquisition and technology acceptance research in the medical setting.
- Developed and analyzed a model of the scheduling workflow using the functional resonance analysis method (FRAM).

- The results of the FRAM were then used by a collaborating research team to inform the development of an optimization algorithm.

Human Factors Researcher, May 2021-August 2023 ~ **Johns Hopkins Hospital**, Baltimore, MD

- Conducted patient safety research for a multi-institutional and multi-disciplinary AHRQ-funded research team.
- Reviewed team projects, developed protocols for IRB approval, led the Johns Hopkins research team, authored two academic research papers (one under review), and presented monthly research updates at collaborator meetings.
- Developed prototypes of implementable devices using Solidworks, conducted usability studies with the technology acceptance model, performed cognitive task analysis, and conducted statistical analysis using multivariate regression.
- Observed over 20 full-length cardiac anesthesia cases in the process of data collection and conducted several in-depth interviews with clinicians using a self-created interview script and interview protocol.
- Designed, managed, and reported an observational study to evaluate the impact of novel interventions in the operating room. Managed the study from start to finish, working with collaborators, instructing a research team, and analyzing data.

UX Research and Design Intern, June 2022-August 2022, June 2023-August 2023. ~ **PTC Inc. (Onshape)**, Boston, MA

- Designed, prototyped, tested, and implemented new functionality for Onshape software using Figma to create prototypes and conducting UX tests to gather user-centered feedback. Prototyping included icon development, workflow creation, webpage redesign, and novel feature ideation and implementation.
- Created a fully functional web application using the Django Python web development framework. Wrote in Python for backend development, and HTML, CSS, and Bootstrap for frontend development.
- Used Python to build a scanning tool for the reading of published ISO standards using Tesseract for optical character recognition and the Pandas package for data frame manipulation and modification.
- Worked with the customer service team using the software I developed to address customer complaints through Zendesk, and worked with developers to integrate the software I developed into the product.

Teaching Experience

Teacher's Assistant, January 2022-December 2023~ **Johns Hopkins University**, Baltimore, MD

- Served as a teaching assistant for Computational Cardiology, Biomedical Data Science, and four semesters of Introduction to Business. Student evaluation methods included quizzes, exams, and written reports.
- Collaborated with the course professors to create evaluation material that appropriately covered content.
- Organized review sessions, provided continued feedback on semester-long projects, and appropriately addressed situations of academic dishonesty and failing performance with the goal of success and learning for all enrolled students.
- Developed a learning environment where students reported feeling supported. With me as the TA, the courses received above-average mid-year scores on a student survey, and all students passed the classes.

First Year Mentor, August 2021-May 2024 ~ **Johns Hopkins University**, Baltimore, MD

- Served as a mentor for incoming students with the goal of helping mentees transition smoothly into college. This year-long position involved orienting my mentees to campus and providing support.
- Helped new students become acclimated to their new life on a college campus and develop community.
- Guided students around campus and provided critical information about campus life, emotional support, and mentorship as the semester progressed; organized a variety of group bonding events to enhance a sense of belonging.
- Mentored students to become academically successful, deeply engaged with campus activities, and confident on campus.

Community Engagement / Volunteer Work

President, September 2025-Present ~ University of Michigan HFES Student Chapter

- Actively serving as the president of the HFES student chapter at the University of Michigan, supporting chapter programming, student engagement, and professional development initiatives.
- Collaborating closely with student and departmental leadership to secure funding and support chapter activities.
- Coordinating with student professional organizations to revitalize the Industrial and Operations Engineering graduate student mentorship program for incoming master's and PhD students.
- Broadening chapter engagement by developing programming for students across human factors-adjacent disciplines, including ergonomics, psychology, computer science, robotics, and statistics.

Student Representative, February 2025-Present ~ HFES Occupational Ergonomics Technical Group

- Actively serving as the student representative on the HFES Occupational Ergonomics Technical Group leadership committee with the goal of providing unique insights from a student's perspective.
- Collaborating closely with the other members of the technical group leadership committee to review conference submissions and promote the research of our technical group members on social media.
- Working to plan a professional development event for students and young professionals meant to highlight the diverse career opportunities and academic backgrounds in occupational ergonomics.

President, September 2020-May 2024 ~ Johns Hopkins University BMES, Baltimore, MD

- Served as the president of the Johns Hopkins Biomedical Engineering Society with the goals of developing coherent marketing messaging, offering more social and professional events, and creating better community engagement.
- Managed the social media accounts for the Biomedical Engineering Society, created and managed the new LinkedIn, oversaw the development of the official BMES website, and worked to curate an official Johns Hopkins BMES brand.
- Worked closely with other board members to revitalize the community outreach program. We are now partnered with public schools and after-school programs to provide instructional sessions about engineering and STEM as a whole.
- The Johns Hopkins BMES has become a well-known student organization with an over 500% increase in membership and an over 100% increase in event attendance.
- Johns Hopkins BMES is now well-connected within the Baltimore community and well-positioned to further its community engagement.

Diversity Admissions Assistant, March 2021-May 2024 ~ Johns Hopkins University Admissions, Baltimore, MD

- Volunteered within the Johns Hopkins University admissions office to improve educational equity
- Worked closely with admissions officers to run tours and host prospective student discussions.
- In a year-long paid position with the Access and Diversity committee (2021-2022), worked tirelessly to improve access to university support systems and resources for prospective students from underrepresented backgrounds.
- In the admissions office, I focused on improving educational equity, and I have had personal conversations with people from many backgrounds providing a personal perspective on higher education admissions processes and attending Johns Hopkins University

Academic Tutor, September 2022-May 2024 ~ SquashWise, Baltimore, MD

- Volunteered as an academic tutor for SquashWise, a Baltimore nonprofit after-school program where local students can get help with their school work while also playing on an elite squash team.
- Tutored a mixture of high school and middle school students, focusing on mathematics and history.
- Worked closely with low-income high school students to successfully apply to top-rated public and private universities

while securing valuable scholarships and applying for need-based financial aid.

- Developed rewarding relationships with all of the students and helped many of my tutees get into their dream schools while also finding the scholarship money to be financially secure.